

Skill Session 1

Task 1

To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

```
j_rakshitha@RAKSHITHA:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

j_rakshitha@RAKSHITHA:~$ git --version
git version 2.53.0
j_rakshitha@RAKSHITHA:~$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
j_rakshitha@RAKSHITHA:~$ mkdir -p ~/ossp_project
j_rakshitha@RAKSHITHA:~$ cd ~/ossp_project
j_rakshitha@RAKSHITHA:~/ossp_project$ mkdir src include build
j_rakshitha@RAKSHITHA:~/ossp_project$ touch src/main.c
j_rakshitha@RAKSHITHA:~/ossp_project$ touch Makefile
j_rakshitha@RAKSHITHA:~/ossp_project$ ls
Makefile  build  include  src
j_rakshitha@RAKSHITHA:~/ossp_project$ tree
.
├── Makefile
├── build
├── include
├── src
│   └── main.c
└── 4 directories, 2 files

j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/main.c
j_rakshitha@RAKSHITHA:~/ossp_project$ cat src/main.c
#include <stdio.h>

int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
j_rakshitha@RAKSHITHA:~/ossp_project$ nano Makefile
j_rakshitha@RAKSHITHA:~/ossp_project$ cat Makefile
```

```
j_rakshitha@RAKSHITHA:~/ossp_project$ cat Makefile
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = build/main
SRC = src/main.c

all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

clean:
    rm -f $(TARGET)

run: all
    ./${TARGET}
j_rakshitha@RAKSHITHA:~/ossp_project$ make
gcc -Wall -Wextra -g src/main.c -o build/main
j_rakshitha@RAKSHITHA:~/ossp_project$ ls build
main
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/main
OSSP Project Started Successfully!
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
GNU nano 8.7.1 src/main.c
#include <stdio.h>

int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
|
```

Task 2

To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing.

```
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/process_demo.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/process_demo.c -o build/process_demo
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/process_demo
Parent Process Started
Parent PID: 1228

Parent waiting for child...

Child Process Created
Child PID: 1229
Parent PID: 1228
Child is executing ls command...
total 16
-rw-r--r-- 1 j_rakshitha j_rakshitha 164 Aug 17 05:32 Makefile
drwxr-xr-x 2 j_rakshitha j_rakshitha 4096 Aug 17 05:39 build
drwxr-xr-x 2 j_rakshitha j_rakshitha 4096 Aug 17 05:28 include
drwxr-xr-x 2 j_rakshitha j_rakshitha 4096 Aug 17 05:39 src
Child process completed.
Parent process exiting.
j_rakshitha@RAKSHITHA:~/ossp_project$ pstree -p
systemd(1)─agetty(238)
            ├chornyd-starter(108)─chornyd(272)─chornyd(282)
            ├cron(109)
            ├dbus-daemon(110)
            └init-systemd(Ub(2))─SessionLeader(725)─Relay(733)(726)─ba+
                                └SessionLeader(939)─Relay(941)(940)─ba+
                                └init(7)─{init}(8)
                                └login(329)─bash(413)
                                └{init-systemd(Ub)}(9)

            networkd-dispat(157)
            rsyslogd(201)─{rsyslogd}(219)
                        └{rsyslogd}(220)
                        └{rsyslogd}(221)
            systemd(383)─(sd-pam)(385)
            systemd-journal(47)
            systemd-logind(168)
            systemd-resolve(82)
            systemd-udev(92)
            unattended-upgr(242)─{unattended-upgr}(295)
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
GNU nano 8.7.1                                     Makefile
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = build/main
SRC = src/main.c

all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

clean:
    rm -f $(TARGET)

run: all
    ./${TARGET}
```

Skill Session 2

Task 1

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
j_rakshitha@RAKSHITHA:~/ossp_project$ ~/ossp_project
-bash: /home/j_rakshitha/ossp_project: Is a directory
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/myshell.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/myshell.c -o build/myshell
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/myshell
Mini Shell Started
Type 'exit' to quit.

mysHELL> exit
Exiting shell...
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
GNU nano 8.7.1 src/myshell.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_INPUT 100

int main() {
    char input[MAX_INPUT];

    printf("Mini Shell Started\n");
    printf("Type 'exit' to quit.\n\n");

    while (1) {
        printf("mysHELL> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL) {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        if (strlen(input) == 0) {
            continue;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

Task 2

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/keyboard_shell.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/keyboard_shell.c -o build/keyboard_shell
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/keyboard_shell
Keyboard Input Shell
Type commands and press Enter.
Use Backspace to edit.
Type 'exit' to quit.

myshell> hey there
Command entered: hey there
myshell> exit
Exiting shell...
j_rakshitha@RAKSHITHA:~/ossp_project$ tree
.
├── Makefile
├── build
│   ├── keyboard_shell
│   ├── main
│   ├── myshell
│   └── process_demo
├── include
└── src
    ├── keyboard_shell.c
    ├── main.c
    ├── myshell.c
    └── process_demo.c

4 directories, 9 files
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX_INPUT 100

void enable_raw_mode(struct termios *original) {
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original) {
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

int main() {
    struct termios original;
    char buffer[MAX_INPUT];
    int position = 0;
    char ch;

    printf("Keyboard Input Shell\n");
    printf("Type commands and press Enter.\n");
    printf("Use Backspace to edit.\n");
    printf("Type 'exit' to quit.\n\n");

    enable_raw_mode(&original);

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        position = 0;

        while (1) {
            if (read(STDIN_FILENO, &ch, 1) <= 0) {
                break;
            }
        }
    }
}
```

Practice Task

Simple Commands

```
j_rakshitha@RAKSHITHA:~$ hostname
RAKSHITHA
j_rakshitha@RAKSHITHA:~$ id
uid=1000(j_rakshitha) gid=1000(j_rakshitha) groups=1000(j_rakshitha),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
j_rakshitha@RAKSHITHA:~$ tty
/dev/pts/0j_rakshitha@RAKSHITHA:~$ uptime
 06:06:20 up 1:19, 1 user, load average: 0.08, 0.02, 0.01
j_rakshitha@RAKSHITHA:~$ users
j_rakshitha@RAKSHITHA:~$ who
j_rakshitha@RAKSHITHA:~$ w
 06:06:35 up 1:19, 1 user, load average: 0.06, 0.02, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
j_rakshi  pts/1    -                04:46    1:19m  0.04s  0.04s  -bash
j_rakshitha@RAKSHITHA:~$ groups
j_rakshitha adm cdrom sudo dip plugdev users
j_rakshitha@RAKSHITHA:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=RAKSHITHA
PWD=/home/j_rakshitha
LOGNAME=j_rakshitha
HOME=/home/j_rakshitha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/2571_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzm=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.t2=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.#=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
```

```
LESSOPEN=| /usr/bin/lesspipe %s
USER=j_rakshitha
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0.x64_8melyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/System32:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/J_Rakshitha/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/J_Rakshitha/AppData/Local/Programs/Python/Python314:/mnt/c/Users/J_Rakshitha/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/J_Rakshitha/.local/bin:/mnt/c/Users/J_Rakshitha/AppData/Local/Microsoft/WindowsApps:/mnt/d/Coding/gcc/bin:/mnt/c/Users/J_Rakshitha/AppData/Roaming/npm:/mnt/d/Coding/vs code/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
_=usr/bin/env
j_rakshitha@RAKSHITHA:~$ printenv
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=RAKSHITHA
PWD=/home/j_rakshitha
LOGNAME=j_rakshitha
HOME=/home/j_rakshitha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/2571_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzm=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.t2=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.#=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
```

```
old=00;90:*_orig=00;90:*_part=00;90:*_rej=00;90:*_swp=00;90:*_tmp=00;90:*_dpgk-dist=00;90:*_dpgk-old=00;90:*_uct-dist=00;90:*_uct-new=00;90:*_uct-old=00;90:
*_rpmnew=00;90:*_rpmorig=00;90:*_rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=j_rakshitha
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCor
poration/1.WindowsSubsystemForLinux_2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Fil
es/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/System32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/m
nt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Fil
es/NVIDIA Corporation/NVIDIA NvDLISR:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/J_Rakshitha/AppData/Local/Pro
grams/Python/Python314/Scripts:/mnt/c/Users/J_Rakshitha/AppData/Local/Programs/Python/Python314:/mnt/c/Users/J_Rakshitha/AppData/Local/Programs/Python/Lau
ncher:/mnt/c/Users/J_Rakshitha/.local/bin:/mnt/c/Users/J_Rakshitha/AppData/Local/Microsoft/WindowsApps:/mnt/d/Coding/gcc/bin:/mnt/c/Users/J_Rakshitha/AppDa
ta/Roaming/npm:/mnt/d/Coding/vs code/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
_=usr/bin/printenv
j_rakshitha@RAKSHITHA:~$
j_rakshitha@RAKSHITHA:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
none            3990856         0    3990856   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3990856         4    3990852   1% /mnt/wsl
drivers         122108924 105095840 17013084  87% /usr/lib/wsl/drivers
/dev/sdd        1055762868  1914348 1000145048   1% /
none            3990856         4    3990812   1% /mnt/wslg
none            3990856         0    3990856   0% /usr/lib/wsl/lib
rootfs          3985096      2772    3982324   1% /init
none            3990856         532    3990324   1% /run
none            3990856         0    3990856   0% /run/lock
none            3990856         0    3990856   0% /run/shm
none            3990856         0    3990776   1% /mnt/wslg/versions.txt
none            3990856         0    3990776   1% /mnt/wslg/doc
C:\             122108924 105095840 17013084  87% /mnt/c
D:\            375665660 48073684 327591976  13% /mnt/d
```

```
tmpfs           3990856         0    3990856   0% /tmp
none            1024         0    1024     0% /run/credentials/systemd-resolved.service
none            1024         0    1024     0% /run/credentials/getty@tty1.service
tmpfs           798168        12    798156   1% /run/user/1000
j_rakshitha@RAKSHITHA:~$ free
              total        used        free      shared  buff/cache   available
Mem:          7981712      582216      7399800        3588       253504       7399496
Swap:         2097152           0       2097152
j_rakshitha@RAKSHITHA:~$ vmstat
procs -----memory-----swap-----io-----system-----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi   bo    in   cs us sy id wa st gu
  0   0     0 7310120    5000 248504     0    0    201   87  110    0  0  0 100  0  0  0
j_rakshitha@RAKSHITHA:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M 1 disk
sdb   8:16   0 159.4M 1 disk
sdc   8:32   0    2G  0 disk [SWAP]
sdd   8:48   0    1T  0 disk /mnt/wslg/distro
j_rakshitha@RAKSHITHA:~$ mount
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft
-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work
,uuid=on)
rootfs on /init type rootfs (ro,size=3985096k,nr_inodes=996274)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3985096k,nr_inodes=996274,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
```

```

cgroup on /sys/fs/cgroup type cgroup (rw,nosuid,nodev,noexec,relatime,nodeviate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs;path=D:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
none on /run/user type tmpfs (rw,relatime)
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,nr_inodes=1048576)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-resolved.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
none on /run/credentials/getty@tty1.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=798168k,nr_inodes=199542,mode=700,uid=1000,gid=1000)
tmpfs on /mnt/wslg/run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=798168k,nr_inodes=199542,mode=700,uid=1000,gid=1000)
j_rakshitha@RAKSHITHA:~$ history
 1 pwd
 2 ls
 3 ls -l
 4 clear
 5 whoami
 6 date
 7 clear
 8 cal
 9 clear
10 hostname
11 uname -a
12 man pwd
13 mkdir OSSP
14 cd OSSP
15 git init
16 mkdir src include obj bin docs tests scripts assets
17 touch .gitignore README.md LICENSE Makefile
18 touch src/main.c,shell.c,parser.c,executor.c,builtin.c}
19 touch include/shell.h,parser.h,executor.h,builtin.h}
20 touch bin/my_shell

```

```

19 touch include/shell.h,parser.h,executor.h,builtin.h}
20 touch bin/my_shell
21 touch docs/{design.md,report.pdf}
22 touch tests/{test_parser.c,test_executor.c}
23 touch scripts/run.sh
24 chmod +x scripts/run.sh
25 touch assets/architecture.png
26 tree
27 sudo apt install tree
28 sudo apt update
29 sudo apt install tree
30 tree
31 touch process.c
32 nano process.c
33 gcc process.c -o process
34 sudo apt update
35 sudo apt install gcc
36 gcc process.c -o process
37 tree
38 uname -a
39 lscpu
40 ps
41 top
42 uname -a
43 lscpu
44 nano command_executor.c
45 ls
46 gcc command_executor.c -o command_executor
47 gcc --version
48 cat command_executor.c
49 sudo apt update
50 sudo apt install build-essential libc6-dev
51 gcc command_executor.c -o command_executor
52 ./command_executor
53 echo $$
54 ps
55 pstree
56 echo $$~
57 echo $$
58 ps -ef
59 pwd

```

```

61 whoami
62 hostname
63 id
64 tty
65 uptime
66 users
67 who
68 w
69 groups
70 env
71 printenv
72 df
73 free
74 vmstat
75 lsblk
76 mount
77 history
78 clear
79 hostname
80 id
81 tty
82 uptime
83 users
84 who
85 w
86 groups
87 env
88 printenv
89 df
90 free
91 vmstat
92 lsblk
93 mount
94 history
j_rakshitha@RAKSHITHA:~$ |

```

```
j_rakshitha@RAKSHITHA:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
j_rakshitha@RAKSHITHA:~$ sudo apt update
sudo apt install sysstat
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [447 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [235 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.4 kB]
Fetched 830 kB in 12s (68.7 kB/s)
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
sysstat is already the newest version (12.7.7-0ubuntu2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 40
j_rakshitha@RAKSHITHA:~$ iostat
Linux 6.18.33.2-microsoft-standard-WSL2 (RAKSHITHA)      08/07/26      _x86_64_      (12 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.31    0.00   0.37   0.09    0.00   99.24

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
sda                1.30       289.37        0.00         0.00     158785         0         0
sdb                0.20        11.81        0.00         0.00       6481         0         0
sdc                0.17         3.59        0.01         0.00       1968         4         0
sdd               19.09       803.46       603.38       906.27    440877    331084    497288
```

```
j_rakshitha@RAKSHITHA:~$ echo "Hello World"
Hello World
j_rakshitha@RAKSHITHA:~$ sudo apt install bc
bc is already the newest version (1.07.1-4build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 40
j_rakshitha@RAKSHITHA:~$ bc
bc 1.07.1
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This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
123456789+987456321
1110913110
quit
j_rakshitha@RAKSHITHA:~$ factor 1234
1234: 2 617
j_rakshitha@RAKSHITHA:~$ |
```

```
y
y
y
y
y
y
y
y
y
y
^C
j_rakshitha@RAKSHITHA:~$ sleep 5
j_rakshitha@RAKSHITHA:~$ |
```